

Multi-Dataset Evaluation of Eye Movement-Based Drowsiness Detection Using ANN and TabNetClassifier

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Abstract—Drowsiness among drivers is one of the major reasons of automobile crash incidents. This study seeks to compare various datasets and create models optimized for utilizing eye movements data in detecting drowsiness. The proposed methodology works by extracting features from eye movements and behavioral data, addressing challenges such as occlusions and computational overhead. Three datasets, NTHU-DDD, UTA-RLDD, and YAW-DD were used in this study. The NTHU-DDD dataset was treated as a binary classification task, while UTA-RLDD and YAW-DD were used for ternary classification. The features were standardized and input into two models: a custom Artificial Neural Network (ANN) and TabNet. The proposed system achieved a peak accuracy of 95.56% on the UTA-RLDD dataset using TabNet, with an AUC of 0.99 and F1-score above 0.92 across classes. These results indicate that the proposed hybrid approach is both accurate and computationally efficient, making it suitable for real-time drowsiness detection in intelligent driver-assistance systems (IDAS).

Index Terms—Driver Drowsiness Detection, Eye-Blink Monitoring, Fatigue Detection Systems, Behavioral analysis, Eye Movements, Transfer Learning, Pupil Dilation

I. INTRODUCTION

Driver fatigue or drowsiness is one of the most common and major causes of road accidents every year. Fatigue can build up while driving for long distances and periods of time. Most drivers have reported that they have experienced fatigue or fallen asleep behind the wheels at least once in their lives. According to a study, more than 50% (3.5–67.3%) of their participants reported that they had been sleep deprived while driving [1].

Driver negligence isn't just caused by fatigue or sleep deprivation; it can also be the result of alcohol consumption, medication, phone use, or other distractions. There are many ongoing researches that explore various means aimed at diminishing such factors or building some solutions to overcome it. Among them, methods usually belong to either physiological, vehicle-based or behavioral measures. Physiological measures have stopped being investigated due to their intrusive nature and expensive costs. Similarly, vehicle-based measures fall

short of being generalizable to all scenarios and vehicles and are costly to setup and maintain. Behavioral measures have found an on-growing demand in research due to recent advancements in computer vision techniques and deep learning [1].

Among these recent research, much of interest is developed over eye-related measures as it is one of the first and clearest signs of distraction, fatigue or drowsiness among drivers. Eye tracking involves monitoring pupils and understanding dynamics of eye movements and related reasons on its basis developing measures to infer about drowsiness of drivers. Some of the methods that involve eye tracking are cited here.

1. Electrooculography (EOG):

This technique measures electrical potentials generated by eye movements using electrodes placed around the eyes [2]. Despite being quite accurate, it is invasive and inappropriate for driving applications in the real world.

2. Infrared (IR) Pupil Tracking:

This method uses infrared light to detect pupil position and eye movements. Cameras track the reflection of IR light on the cornea and retina, providing precise gaze direction. Because of their accuracy and dependability in low light, infrared-based techniques are widely used.

3. Video-Based Tracking:

This method employs standard or infrared cameras which are used to capture images of drivers' eyes. And then uses computer vision algorithms to track eye movements, blinks, and gaze direction. Since it is inexpensive and non-intrusive, video-based tracking is preferred for real-world uses like driver monitoring.

4. Gaze Estimation via Deep Learning:

Recent advancements leverage deep learning models to analyze video feeds for gaze estimation. Convolutional Neural Networks (CNNs) are used to identify eye landmarks, track pupil movements, and determine gaze direction. These techniques combine a high degree of accuracy with flexibility in a range of environmental conditions [3].

Eye movement analysis using deep learning has emerged as one of the pivotal topics in current research in driver drowsiness. It is seen as a reliable indicator of drowsiness, fatigue or distraction [3]. Studies have shown facial landmarks are the best identifiers of drowsiness. This along with other reliable measures like Mouth aspect ratio to measure yawning or temporal feature like blinking and head tilt can easily define various classes of drowsiness and distraction [4].

This study builds upon these advancements. So, the objectives of this study are:

- To develop an accurate and lightweight deep learning model for detecting driver fatigue using eye movement features such as EAR, MAR, and pupil coordinates.
- To evaluate and compare the performance of ANN and TabNet classifiers across multiple datasets for real-time fatigue detection.

II. LITERATURE REVIEW

Drowsiness detection systems have become the key point of road safety research due to its ability to reduce fatigue-related accidents. Recent developments that have happened in the domain of deep learning and computer vision-based methods along with sensor systems have prompted advancements in the field of drowsiness detection.

Alguindigue and Jose (2023) [5] had studied a machine learning-based approach for driver drowsiness detection, which emphasized on eye-tracking techniques along with heart rate variability (HRV) as well as electro-dermal activity (EDA). Their study formed three models of which the eye based model that learned from spatial features such as Eye Aspect Ratio (EAR) and gaze direction to assess drowsiness levels achieved a 90% accuracy. However, using multiple data sources increased computational overhead, hindering real-time deployment.

Albadawi et al. (2023) [6] have developed a real-time drowsiness detection system which was based on image-based features such as EAR, MAR, and head pose using CV methods. They achieved nearly 99% accuracy on the NTHU-DDD dataset by feeding these features into classifiers like Random Forest, Neural Networks, and SVM. However, factors like camera quality, lighting conditions, and occlusions affected the performance of their solution.

Florez et al. (2024) [7] studied an approach based on real-time embedded system which leveraged a custom deep learning model, DD-AI CNN for detecting driver drowsiness. During controlled trials, the model reached a remarkable accuracy of 99.88%. However, when deployed in a practical setting using Jetson Nano hardware within a vehicle, the accuracy decreased marginally to 96.55%, yet remained exceptionally effective. However, the performance is reliant on hardware-specific optimizations.

Lian et al. (2024) [8] have studied a hybrid system that integrates Electroencephalography (EEG) and eye behavior to monitor driver fatigue to enhance performance and improve its reliability. They leveraged EEG signals for observation of neural activity and analyzed gaze and eyelid movements using

eye-tracking. The study, tested on real-world driving scenarios, achieved an average accuracies of 97% in intra-session but 85% in cross-session and 75.4% in cross-subject session. Their results are subject to variability due to adjustments in the eye-tracking time window.

Essahraoui et al. (2025) [9] have studied K-Nearest Neighbors (KNN) and advanced CV models namely YOLOv5 and YOLOv8 for real-time driver monitoring systems and used facial analysis. Their system achieved an accuracy of 98.89% with KNN and 100% precision and recall with YOLOv5 and YOLOv8 on the UTA-RLDD dataset. The study acknowledged that the limited dataset and focus on individual features constrained the generalization of their findings.

Wang et al. (2024) [10] have proposed a multi-task Driver Monitoring System (VDMoE) that combines RGB video with remote photoplethysmography (rPPG) to integrate facial and physiological features. Their system demonstrated an improvement in drowsiness detection and cognitive load by optimizing Mixture-of-Experts (MoE) structures. Yet, the over-reliance on multimodal data introduced a demand in additional processing complexity and thus an increased demand for computational resources.

Pradeep et al. (2023) [11] have studied utilization of Support Vector Machine (SVM) in developing a real-time system to detect driver fatigue. Their approach involves extracting key features such as EAR, MOR, and Nose Length Ratio using facial landmarks detected from recorded video. They employed adaptive thresholding techniques to identify drowsy states with a sensitivity of 95.58%. Despite the system's high sensitivity, its offline execution of machine learning models deters its use in real-time applications.

Anoop and Deivanathan (2023) [12] have presented a deep learning-based model that detects drowsiness in drivers in extreme conditions such as low light. They gave special importance to classifying driver states—normal, eye closed, and yawning—by analyzing facial features captured from multiple camera angles allowing to develop a multi-class data. The model has achieved a classification accuracy of 98.3%. The study's over-reliance on well positioning of camera and low light optimization suggests that the performance is still sensitive to external factors and there is a need for further exploration of advanced feature extraction techniques.

Pauly et al. (2023) [13] also employed Support Vector Machines, along with Histogram of Oriented Gradients (HOG). It achieved an accuracy of 91.6% in detecting drowsiness in real-time. Although the efficacy of the model can be compromised under varying head tilts and lighting conditions, its real-time capabilities mark its importance for early alert systems.

The literature shows that transfer learning and pre-trained models can be used to improve adaptability and efficiency in fatigue detection systems. These techniques enable more accurate and scalable driver monitoring solutions. Our study aims to use these advances by fine-tuning deep learning models on eye movement behavior data, addressing challenges such as environmental variability and computational overhead. The review of previous studies is discussed in Table 1.

III. METHODOLOGY

The proposed methodology initially employed pretrained models such as VGG-16, ResNet50, and MobileNetV2, but these yielded suboptimal accuracy between 50% and 70%. Consequently, the approach was revised to utilize feature extraction combined with a custom neural network and TabNet.

A. Dataset Collection and Preparation

Three datasets were collected, namely, National Tsing Hua University Drowsy Driver Detection (NTHU-DDD), University of Texas at Arlington Real-Life Drowsiness Dataset (UTA-RLDD) and Yawning Detection Dataset (YAW-DD). Of these, NTHU-DDD is an image-based binary dataset with classes 'drowsy' and 'alert', while UTA-RLDD and YAW-DD are video-based ternary-class datasets. UTARLDD consisted of classes 'alert', 'low vigilant' and 'drowsy' and YAWDD consisted of 'Normal', 'Talking' and 'Yawning' which can be mapped to those of UTARLDD.

B. Feature Extraction

The images and videos were preprocessed so that key facial landmarks could be extracted using MediaPipe's 'FaceMesh' module, which captures 468 facial landmarks from the 2D images gathered from the datasets as shown in Fig. 1, from which these features were extracted:

1. Pupil Coordinates: Using facial landmark, the coordinates of the pupils were mapped to that of screen coordinates.
2. Eye Aspect Ratio (EAR): An indicator of eye openness, calculated using specific eye landmarks with the formula in eq. (1) [14].

$$EAR = \frac{|F3 - F4| + |F5 - F6|}{2 \times |F2 - F1|} \quad (1)$$

where $F1, F2, F3, F4, F5, F6$ are the eye landmarks.

They are:

- i. F1: Outer corner of the eye
- ii. F2: Inner corner of the eye
- iii. F3: Upper middle
- iv. F4: Upper edge
- v. F5: Lower middle
- vi. F6: Lower edge

3. Mouth Aspect Ratio (MAR): Captures mouth movements, specifically yawning, by analyzing distances between mouth landmarks with eq. (2) [15].

$$MAR = \frac{|M3 - M4| + |M5 - M6|}{2 \times |M1 - M2|} \quad (2)$$

where $M1, M2, M3, M4, M5, M6$ are the mouth landmarks.

They are:

- i. M1, M2: Mouth width (horizontal edges)
- ii. M3, M4: Upper and lower midpoints
- iii. M5, M6: Secondary vertical midpoints

TABLE I
REVIEW OF PREVIOUS STUDIES ON DRIVER DROWSINESS DETECTION

Ref.	Methodology	Dataset Used	Classification Type	Findings	Challenges
[5]	Eye-tracking, HRV, EDA	Video-based (40 participants)	Binary	A: 90% (Eye-based model)	High computational overhead due to multiple data sources
[6]	EAR, MAR, Head Pose (Random Forest, SVM, Neural Networks)	NTHU-DDD	Binary	A: 99%	Affected by occlusions, camera quality, and lighting
[7]	Custom CNN (DD-AI CNN)	NITYMED	Binary	A: 99.88% (Controlled trials)	Hardware-specific optimizations required for real-time performance
[8]	EEG + Eye-tracking Hybrid System	Hybrid dataset (14 participants)	Binary	A: 97% (Intra-session)	Variability due to eye-tracking time window adjustments
[9]	Facial Analysis (KNN, YOLOv5, YOLOv8)	UTA-RLDD, NTHUDD, YAWDD	Binary	A: 98.89% (KNN), 100% (YOLOv5, YOLOv8)	Limited dataset and feature dependency
[10]	Multimodal (RGB + rPPG, MoE)	MCDD	-	A: Not specified	Increased computational demand due to multimodal data processing
[11]	SVM with EAR, MOR, NLR	Web cam-based	Binary	S: 95.58%	Offline execution limits real-time use
[12]	Deep Learning for Multi-Class Classification (Eye Closed, Yawning)	Web cam-based	Multi-class	A: 98.3%	Sensitive to camera positioning, low-light optimization required
[13]	HOG + SVM	Video-based (12 participants)	Binary	A: 91.6%	Affected by head tilt variations and lighting conditions

* A - Accuracy, S - Specificity

For each image, a feature vector was formed by concatenating them into an array.

C. Feature Scaling

To ensure proper normalization and improve the model's training efficiency, feature scaling was applied. The standardized features have mean set to 0 and their standard deviation becomes 1 for which the 'StandardScaler' module was used. The model's performance is enhanced due to equal contribution from all features to the learning process.



Fig. 1. Data Preprocessing

D. Data Splitting

The features were split into train and validation sets with a ratio of 8:2 over whole dataset. The test sets were used to further tune in the models while training and finally the evaluations for result metrics were taken over randomized test sets split from the three datasets.

E. Model Architecture

Two architectures were trained over the three features sets extracted from the three datasets respectively.

First architecture (Hybrid ANN Model) :

1. Input Layer: One fully connected or dense layer of 128 neurons and ReLU activation, accepting 7 input features.
2. Hidden Layers: Two additional Dense layers with 128 and 64 neurons, both using ReLU activation.
3. Dropout Layers: Each hidden layer is followed by a Dropout (0.3) to prevent overfitting.
4. Output Layer: A Dense layer of 3 neurons and softmax activation to do multi-class classification.

Second architecture (TabNet Classifier):

TabNet is a deep learning architecture which was made for tabular data and based on self-attention and feature selection mechanisms to dynamically decide which features to focus on during training. It was configured to explicitly run over CUDA (GPU) for accelerated training.

The methodology has been explained in Fig. 2.

F. Training

The hybrid neural network has been trained on the Adam optimizer with a learning rate of 0.001 and the loss function used was sparse categorical cross-entropy. The models were trained over 20 epochs with a batch size of 32 and then validation data was used to monitor performance. The model was optimized to minimize classification loss while maximizing accuracy.

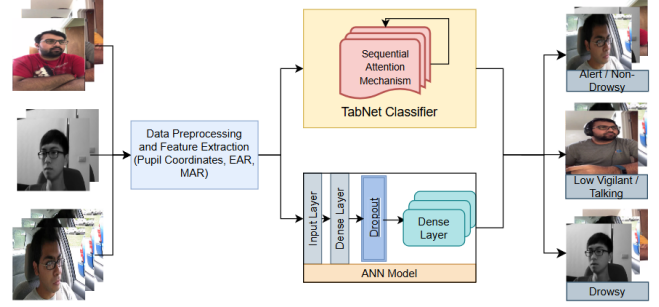


Fig. 2. Framework for Driver Drowsiness Detection

The TabNetClassifier also used an Adam optimizer over a learning rate of 0.02 for 10 epochs. It used early stopping with a patience of 10 to prevent overfitting, and training was performed in batches of 256 with a virtual batch size of 128. The sparse categorical cross-entropy loss is calculated using eq. (4).

$$L = -\frac{1}{N} \sum_{i=1}^N \log P(y_i | x_i) \quad (3)$$

where:

- N is the number of samples.
- y_i is the true class label (an integer).
- $P(y_i | x_i)$ is the predicted probability of the true class.

G. Evaluation Metrics

- Accuracy: It measures the correctly classified samples as a ratio of all classified images and was tracked during training and final evaluation.
- Confusion Matrix: Matrix to provide insights into model's classification of data in test set. Easy to track misclassifications.
- ROC and AUC: The Receiver Operating Characteristic (ROC) curve was plotted to analyze the trade-off between detecting drowsiness and non-drowsiness [16]. Whereas AUC was calculated to get a single numerical measure of model performance.
- Classification Report: A detailed report including precision, recall, and F1-score, which provides a better understanding of the model's ability to distinguish between the classes [17].

IV. RESULTS AND DISCUSSIONS

The results can be mainly compared between the two model for the three datasets. First of all a comparison of their accuracy and f1-score has been tabulated in table 2 and 3. Here, class 0 represents 'alert', class 1 represents 'less vigilant' and class 2 represents 'drowsy' in datasets YAWDD and UTARLDD in table 3 whereas class 0 represents 'alert' and class 1 represents 'drowsy' in NTHUDDD dataset in table 2.

On the NTHU-DDD dataset, ANN outperformed TabNet with higher accuracy (92.89% vs. 88.16%) and better F1-scores across both classes. For YAW-DD, ANN achieved

slightly better accuracy (79.85% vs. 77.91%), but TabNet performed better on class 2 (0.76% vs. 0.72%). On the UTA-RLDD dataset, TabNet clearly outperformed ANN in both accuracy (95.56% vs. 91.15%) and F1-scores across all three classes.

TABLE II
FINDINGS OF ANN AND TABNET MODELS ON NTHUDD DATASET

Dataset	Model	Accuracy	F1-score (0)	F1-score (1)
NTHUDD	ANN	0.9289	0.92	0.94
	TabNet	0.8816	0.87	0.89

TABLE III
FINDINGS OF ANN AND TABNET MODELS ON OTHER DATASETS

Dataset	Model	Accuracy	F1-score (0)	F1-score (1)	F1-score (2)
YAWDD	ANN	0.7985	0.81	0.83	0.72
	TabNet	0.7791	0.82	0.76	0.76
UTARLDD	ANN	0.9115	0.91	0.90	0.92
	TabNet	0.9556	0.95	0.95	0.96

A. Confusion Matrix

- Confusion Matrix: Heatmap was used to display the confusion matrices (Fig. 3-5). For each dataset the confusion matrices have been shown to compare results between the hybrid ANN model and TabNet model.

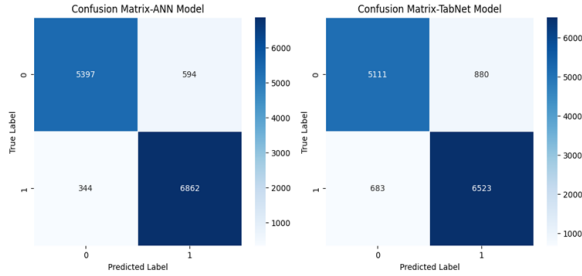


Fig. 3. Confusion matrices - NTHUDD

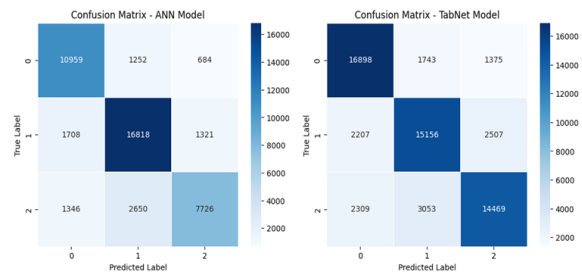


Fig. 4. Confusion matrices - YAWDD

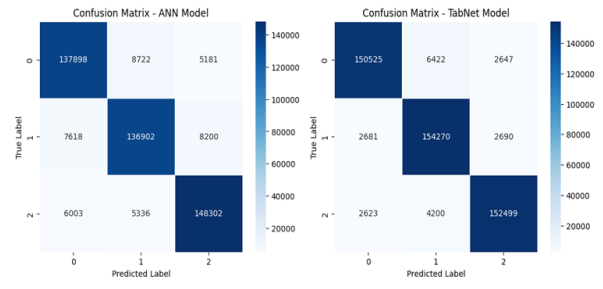


Fig. 5. Confusion matrices - UTARLDD

B. ROC Curve

- ROC Curve: The curves for each dataset were plotted in Fig. 6-8 to display the performance trade-offs, along with the AUC (Area Under the Curve) score for overall performance evaluation.

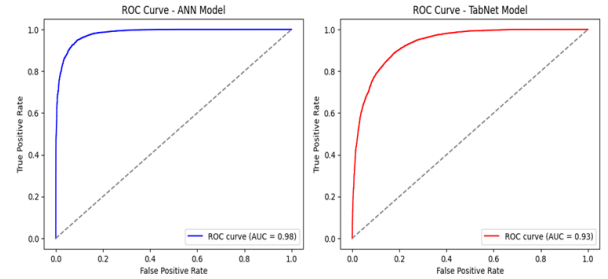


Fig. 6. ROC Curves - NTHUDD

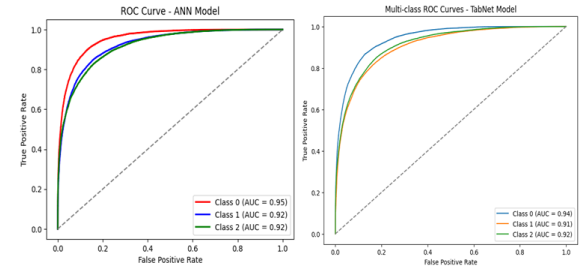


Fig. 7. ROC Curves - YAWDD

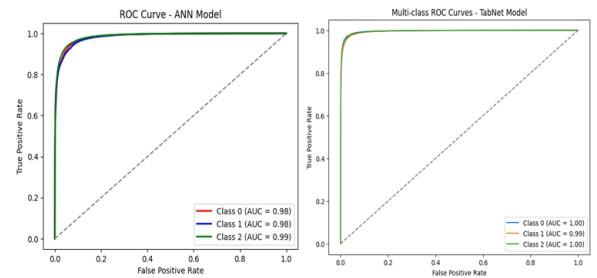


Fig. 8. ROC Curves - UTARLDD

The above results show that models found it hard to train on YAWDD dataset. This lies in reason that the 'talking' class provided tends to be misclassified either to alert or drowsy as MAR as a feature changes to match drowsy state whereas EAR and eye coordinates resemble the alert state. Whereas the models showed better results in other two datasets, pointing out the importance of Pupil coordinates, EAR and MAR as a feature in driver drowsiness detection. Each component used for feature extraction augments the feature set, enabling the model to identify intricate patterns in the data, thereby enhancing accuracy and overall performance.

V. CONCLUSION AND FUTURE WORK

This study demonstrates that fundamental facial landmarks can be used as a starting point for extracting attributes like pupil coordinates, EAR, and MAR. The proposed system demonstrated strong performance on the UTARLDD dataset, achieving a peak accuracy of 95.56%, an AUC of 0.99, and an F1-score exceeding 0.92 across all classes. This approach reduces the computational load compared to employing spatial and temporal features from images when developing a model for detecting drowsiness. Moreover, it characterizes features that are prone to overlapping classes but remain unaffected by occlusions and lighting issues, provided that the facial mesh is accurately identified. The research further indicates that TabNet serves as an effective model for detecting drowsiness, offering advantages over the approach of employing pretrained CNN models for feature extraction followed by classification. Our models exhibit notable accuracy and impressive recall; however, they encounter hurdles, as the extracted features continue to be influenced by positioning. To mitigate the variations that occur in overlapping classes, integrating additional features such as head pose and nose angle might be beneficial. Further research is necessary in spatial features to improve and enhance reliability.

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